12311410 [illegible] Real Analysis (H) Midterm.

1. Simplest example: $K=\{a,b\}$, where $a=(0,0)$, $b=(0,1)$ $\Rightarrow$ bdd & closed $\Rightarrow$ cpt.

$$f: K \to K \quad \Rightarrow \text{ no fixed pts.}$$
$$a \mapsto b$$
$$b \mapsto a.$$

Why contracting mapping principle is not applicable?

f is not contracting! $\quad \rho(f(x),f(y)) = \begin{cases} 0, & x=y \\ 1, & x\neq y \end{cases}$

$\Rightarrow \nexists\, q\in[0,1)$, s.t. $\rho(f(x),f(y)) \le q\cdot\rho(x,y)$, $\forall x,y\in X$.

2. NO.

$f(x+y)=f(x)+f(y)\ \forall x,y\in\mathbb{R}$.

$f(0+0)=f(0)+f(0) \Rightarrow f(0)=0$.

$f(x+x)=f(x)+f(x) \Rightarrow f(2x)=2f(x)$.

$f(x+2x)=f(x)+f(2x) \Rightarrow f(3x)=3f(x)$.

.....

by induction, $f(nx)=n f(x)$, $\forall n\in\mathbb{N}$.

$f(-x+x)=f(-x)+f(x) \Rightarrow f(x)=-f(-x)$.

$\Rightarrow f(nx)=n f(x)$, $\forall n\in\mathbb{Z}$.

$\Rightarrow f(n)=n f(1)$, $\forall n\in\mathbb{Z}$.

$\forall y\in\mathbb{Q}$, write $y=\frac{q}{p}$, $p,q\in\mathbb{Z}$, $p>0$,

$\Rightarrow \underset{\substack{\parallel \\ q\cdot f(1)}}{f(q)} = f(p\cdot\frac{q}{p}) = p\cdot f(\frac{q}{p}) \Rightarrow f(\frac{q}{p})=\frac{q}{p}\cdot f(1)$.

$\Rightarrow f(\frac{q}{p}\cdot x)=\frac{1}{p}\cdot p\, f(\frac{q}{p}\cdot x)=\frac{1}{p} f(qx)=\frac{q}{p} f(x)$. $\forall x\in\mathbb{R}$.

$\Rightarrow f(x+y)=f(x)+f(y)$. $\forall x,y\in\mathbb{R}$

$f(\frac{q}{p}x)=\frac{q}{p}f(x)$. $\forall x,y\in\mathbb{R}$, $\frac{q}{p}\in\mathbb{Q}$

$\Rightarrow f$ is $\mathbb{Q}$-linear.

Consider $\mathbb{R}$ as a linear sp. over $\mathbb{Q}$. ($\mathbb{R}$ is a separable metric sp with dense $\mathbb{Q}\subset\mathbb{R}$). countable

$\forall x\in\mathbb{R}$, $x=\sum_{i=1}^{n} q_i e_i$, with $q_i\in\mathbb{Q}$, and $\{e_1,e_2,\cdots\}$ is a countable basis (existence guaranteed in class).

Construct f which is $\mathbb{Q}$-linear: $f(x)=f(\sum_{i=1}^{n} q_i e_i)=\sum_{i=1}^{n} q_i f(e_i)$ Such f satisfies $f(x+y)=f(x)+f(y)$.

Now, firstly choose $e_1 = 1$, $e_2 = \sqrt{2}$, $f(1) \triangleq \sqrt{2}$, $f(\sqrt{2}) \triangleq 1$.

If f is linear, since $f(\frac{q}{p}) = \frac{q}{p} f(1)$ for $\frac{q}{p} \in \mathbb{Q}$, $f(x) = f(1)x$ for $x \in \mathbb{R}$.

But here, $f(\sqrt{2}) = 1 \neq 2 = \sqrt{2} \cdot f(1)$.

We can have f satisfying $f(x+y) = f(x) + f(y)$ but violating $f(x) = x \cdot f(1)$!

3. <u>Proof</u>: Firstly, to avoid things like $1 = 0.\dot{9}$, we always choose the decimal expansion that does not end with an infinite string of 9s.

We approximate A by "first n digits" sets.

$\forall n \in \mathbb{N}$, define $A_n := \{ x \in [0,1] : \text{none of the first } n \text{ digits of } x \text{ is } 5 \}$.

$\Rightarrow$ Clearly, $A_1 \supset A_2 \supset \cdots \supset A_n \supset \cdots$

and $A = \bigcap_{n=1}^{\infty} A_n$.

For each A_n, $A_n = \bigcup_{d_1, \cdots, d_n \in \{0,1,2,3,4,6,7,8,9\}} \left[\frac{d_1}{10} + \frac{d_2}{10^2} + \cdots + \frac{d_n}{10^n}, \frac{d_1}{10} + \frac{d_2}{10^2} + \cdots + \frac{d_n}{10^n} + \frac{1}{10^n} \right)$.

$$=: \bigcup_{d_1, \cdots, d_n \in \{0,1,2,3,4,6,7,8,9\}} I_{d_1, \cdots, d_n}.$$

$\forall A_n$ is a finite union of Borel sets in $\mathbb{R}$, so A_n is Borel, hence Lebesgue measurable.

$$\mu(A_n) = \underbrace{9^n}_{\substack{\#\text{ of choices} \\ \text{of } d_1, \cdots, d_n}} \times \underbrace{\frac{1}{10^n}}_{\text{length of } I_{d_1, \cdots, d_n}} = \left(\frac{9}{10}\right)^n.$$

By continuity of measure, $\mu(A) = \mu(\bigcap_{n=1}^{\infty} A_n) = \lim_{n \to \infty} \mu(A_n) = \lim_{n \to \infty} \left(\frac{9}{10}\right)^n = 0$. $\square$

4. <u>Proof</u>: We firstly need to ensure $\mathcal{A} := \{ A \subset \Omega : A \cap \Omega_j \in \mathcal{A}_j \}$ is a σ-algebra.

① $\Omega \cap \Omega_j = \Omega_j \in \mathcal{A}_j$ since $\mathcal{A}_j$ is a σ-algebra.

$\Rightarrow \Omega \in \mathcal{A}$. ✓

② $\forall A \in \mathcal{A}$, $(\Omega \setminus A) \cap \Omega_j = (\Omega \cap \Omega_j) \setminus (A \cap \Omega_j)$

$$= \Omega_j \setminus (A \cap \Omega_j)$$

$A \in \mathcal{A} \Rightarrow A \cap \Omega_j \in \mathcal{A}_j \Rightarrow \Omega_j \setminus (A \cap \Omega_j) \in \mathcal{A}_j. \Rightarrow (\Omega \setminus A) \cap \Omega_j \in \mathcal{A}.$ ✓

③ For $A = \bigcup_{k=1}^{\infty} A_k$, with $\forall A_k \in \mathcal{A}$.

$$\left(\bigcup_{k=1}^{\infty} A_k\right) \cap \Omega_j = \bigcup_{k=1}^{\infty} \underbrace{(A_k \cap \Omega_j)}_{\in \mathcal{A}_j} \in \mathcal{A}_j. \text{ since } \mathcal{A}_j \text{ is a } \sigma\text{-algebra.}$$

$$\Rightarrow \bigcup_{k=1}^{\infty} A_k \in \mathcal{A}. \checkmark$$

Thus, $\mathcal{A}$ is a σ-algebra.

Next, we show μ is a σ-additive measure on $\mathcal{A}$.

$$\mu(A) = \sum_{j=1}^{\infty} \mu_j (A \cap \Omega_j).$$

① $\forall A$, $\mu(A) \geq 0$ since $\mu_j(A \cap \Omega_j) \geq 0$, $\forall j$. ✓

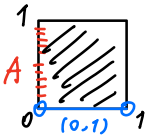
② $\forall A = \bigsqcup_{k=1}^{\infty} A_k$ with $A_k \in \mathcal{A}$.

$$\mu(A) = \mu\left(\bigsqcup_{k=1}^{\infty} A_k\right) \underset{\text{def}}{=} \sum_{j=1}^{\infty} \mu_j\left(\left(\bigsqcup_{k=1}^{\infty} A_k\right) \cap \Omega_j\right) = \sum_{j=1}^{\infty} \mu_j\left(\bigsqcup_{k=1}^{\infty} (A_k \cap \Omega_j)\right).$$

$$\underset{\sigma\text{-additivity of } \mu_j}{=} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \mu_j (A_k \cap \Omega_j) = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \mu_j (A_k \cap \Omega_j) \underset{\text{def}}{=} \sum_{k=1}^{\infty} \mu(A_k). \checkmark$$

Thus, μ is a σ-additive measure on $\mathcal{A}$. $\square$.

5. We know that $\exists A \subset [0,1]$, A- non-measurable.



Define $B := A \times (0,1) \subset [0,1] \times [0,1]$, with projections $\pi_1(B) = \underset{\text{non-measurable}}{A}$, $\pi_2(B) = \underset{\text{measurable}}{(0,1)}$.

Clearly, B is not measurable, otherwise $\pi_1(B) = A$ is measurable.

(General fact: projections of Lebesgue measurable subsets of $\mathbb{R}^2$ are measurable.)

Then, let $Q := [0,1] \times [0,1]$. $\partial Q = (\{0,1\} \times [0,1]) \cup ([0,1] \times \{0,1\})$.

Define $C := B \cup \partial Q$.

Still, C is not measurable. otherwise: $B = C \setminus \partial Q$. ∂Q has measure 0

$\Rightarrow B$ is measurable. ↯.

However, $\pi_1(C) = [0,1]$ and $\pi_2(C) = [0,1]$, which are both measurable.

So, C is what we desire for.

6. <u>Proof</u>: ① For $h > 0$, $g(t+h) - g(t)$

$$= \mu\{x \in \Omega : f(x) \le t+h\} - \mu\{x \in \Omega : f(x) \le t\} \ge 0, \quad ✓$$

since $\{x \in \Omega : f(x) \le t+h\} \supseteq \{x \in \Omega : f(x) \le t\}$

and μ has monotonicity.

② $\forall t \in \mathbb{R}$. WTS: $\lim_{h \to 0^+} g(t+h) = g(t)$.

why? $\Leftarrow$ $\forall \{t_n\}_{n=1}^{\infty}$ s.t. $t_n \downarrow t$ $(n \to \infty)$, we have $\lim_{n\to\infty} g(t_n) = g(t)$.

$\forall \{t_n\}$: $t_n \downarrow t$ $(n \to \infty)$. $E_n := \{x \in \Omega : f(x) \le t_n\}$.

$\Rightarrow E_1 \supset E_2 \supset \cdots \supset E_n \supset \cdots$. and $\underbrace{\bigcap_{n=1}^{\infty} E_n = \{x \in \Omega : f(x) \le t\}}$. $(*)$

$(*)$: "$\subseteq$": $\forall x \in \bigcap_{n=1}^{\infty} E_n$, $f(x) \le t_n$. let $n \to \infty$, $f(x) \le t$. ✓

"$\supseteq$": $\forall x \in \{x \in \Omega : f(x) \le t\}$, $f(x) \le t \le t_n$, $\forall n \Rightarrow x \in \bigcap_{n=1}^{\infty} E_n$. ✓

by continuity of μ, $g(t) = \mu(\bigcap_{n=1}^{\infty} E_n) = \lim_{n\to\infty} \mu(E_n) = \lim_{n\to\infty} g(t_n)$. ✓.

③ Left-continuity:

$\forall \{t_n\}$: $t_n \uparrow t$ $(n \to \infty)$, $E_n := \{x \in \Omega : f(x) \le t_n\}$.

$\Rightarrow E_1 \subset E_2 \subset \cdots \subset E_n \subset \cdots$. and $\underbrace{\bigcup_{n=1}^{\infty} E_n \supsetneq \{x \in \Omega : f(x) \le t\}}$. $(**)$

(**) may not hold.

For example, $t_n = t - \frac{1}{n}$. $t_n \uparrow t$ $(n \to \infty)$ but $t_n \neq t$ $\forall n$.

$\Rightarrow \bigcup_{n=1}^{\infty} E_n = \{x \in \Omega : f(x) < t\}$.

"$\subseteq$": $\forall x \in \bigcup_{n=1}^{\infty} E_n$, $f(x) \le t_n = t - \frac{1}{n} < t$ for some n. ✓.

"$\supseteq$": $\forall x \in \{x \in \Omega : f(x) < t\}$, $\exists n$, s.t. $f(x) \le t - \frac{1}{n}$. $\Rightarrow x \in \bigcup_{n=1}^{\infty} E_n$. ✓.

by continuity of μ, $\lim_{n\to\infty} g(t_n) = \mu(\bigcap_{n=1}^{\infty} E_n) = \mu\{x \in \Omega : f(x) < t\} \overset{?}{=} g(t)$

"=" holds iff $\mu\{x \in \Omega : f(x) = t\} = 0$.

This may be wrong if we choose: $\Omega = \{a\}$. $\mathcal{A} = \{\emptyset, \{a\}\}$. $\mu(\{a\}) = 1$.

$f : \Omega \to \mathbb{R}$, $a \mapsto 0$. $\quad g(t) = \mu\{x \in \Omega : f(x) \le t\} = \begin{cases} \mu(\emptyset) = 0 & \text{if } t < 0. \\ \mu\{a\} = 1 & \text{if } t \ge 0. \end{cases}$

$\Rightarrow g(t)$ is right-continuous but not left continuous at $t = 0$. $\square$.

8pts

7. In class, we have had a non-Borel measurable set in C_0 which is the standard Cantor set. Denote it as A.

Simply consider the set $E := A \times \{0\} \subset \mathbb{R}^2$.

There is a mapping $f : \mathbb{R} \to \mathbb{R}^2$, $x \mapsto (x, 0)$. $\quad f^{-1}(E) = A$.

f is clearly a continuous map. $\Rightarrow E$ is non-Borel, otherwise $A = f^{-1}(E)$ is Borel.

Also, E is obviously measurable in $\mathbb{R}^2$, since $E \subset C_0 \times \{0\}$, which has measure 0.

$\Rightarrow E$ is what we desire for.

10pts

8. ① $s \in [0, 1)$.

Objective: $U \subset [0, 1]$, $\partial U = \bar{U} \setminus U$, $\mu(\partial U) = s$.

$[0, 1] \setminus U = [0, 1] \setminus (\bar{U} \setminus \partial U)$.

Simplification: want $\bar{U} = [0, 1]$, so that $[0, 1] \setminus U = \partial U$.

Let $K := [0,1] \setminus U$. K: closed. $\mu(K) = s$.

$U = [0,1] \setminus K$. $\bar{U} = [0,1] \Rightarrow K$: nowhere dence.

objective: build K: nowhere dense set of measure s.

We construct a fat Cantor set.

step 0: $E_0 = [0,1]$. → left with $1 \cdot s^{\frac{1}{2}}$ measure

step 1: cut off an open interval of length $\left[1 - s^{(\frac{1}{2})}\right] \cdot |E_0|$

$E_1 = E_0 \setminus \left(\frac{1}{2} s^{\frac{1}{2}}, 1 - \frac{1}{2} s^{\frac{1}{2}}\right)$. $\mu(E_1) = 1 \cdot s^{\frac{1}{2}}$

step 2: E_1 has two closed subinterval. → left with $1 \cdot s^{\frac{1}{2}} \cdot s^{\frac{1}{4}}$ measure.

cut off an open interval of length $\frac{1}{2}\left[1 - s^{(\frac{1}{2})}\right] \cdot |E_1|$ separately at the middle.

$\mu(E_2) = 1 \cdot s^{\frac{1}{2}} \cdot s^{\frac{1}{4}}$.

$\vdots$

step k: $\mu(E_k) = 1 \cdot s^{\frac{1}{2}} \cdot \cdots \cdot s^{\frac{1}{2^k}}$. $E_0 \supset E_1 \supset \cdots \supset E_k \supset \cdots$.

Let $K = \bigcap_{k=0}^{\infty} E_k$. $\mu(K) = \mu\left(\bigcap_{k=0}^{\infty} E_k\right) = \lim_{k \to \infty} \mu(E_k) = \lim_{k \to \infty} s^{\left(\frac{1}{2} + \frac{1}{4} + \cdots \frac{1}{2^k}\right)} = s$.

$U = [0,1] \setminus K$ is what we desire for.

② $s = 1$.

Suppose it's possible to have $U \subset [0,1]$, $\partial U = \bar{U} \setminus U$, $\mu(\partial U) = 1$.

$\Rightarrow \mu(\bar{U} \setminus U) = 1$. i.e. $\mu(\bar{U}) - \mu(U) = 1$.

but $0 \le \mu(\bar{U}), \mu(U) \le 1$. $\Rightarrow \mu(\bar{U}) = 1$, $\mu(U) = 0$.

U is a measure-zero open set in $[0,1]$.

If $U \neq \emptyset$, i.e. $\exists x \in [0,1]$, $x \in U$. U open $\Rightarrow \exists B_r(x) \subset U$. $\Rightarrow \mu(U) \ge 2r > 0$. ↯.

$\Rightarrow U = \emptyset$. $\Rightarrow \bar{U} = \emptyset$. but $\mu(\bar{U}) = 1$. ↯.

$\Rightarrow$ it's impossible to construct an open $U \subset [0,1]$, with $\mu(\partial U) = 1$.

10pts

9. No.

Let $c(x)$ be the Cantor function, which is continuous.

Let $g(x) := c(x) + x$. g is a bijection from $[0,1]$ to $[0,2]$. g, g^{-1}: both continuous.

Let C_0 be the Cantor set. $\Rightarrow \mu(g(C_0)) = 1$.

Inside $g(C_0)$ choose a non-measurable set, e.g. a Vitali-style subset,

namely $D \subset g(C_0)$.

completeness

$\Rightarrow g^{-1}(D) \subset C_0$. $\Rightarrow g^{-1}(D)$ is measurable and $\mu(g^{-1}(D)) = 0$.

Let $f = 1_{g^{-1}(D)}(x)$ – the characteristic function : Lebesgue measurable.

$$(f \circ g^{-1})(y) = 1_{g^{-1}(D)}(g^{-1}(y)) = \underbrace{1_D(y)}.$$

char. func. of D, which is non-measurable.

$\Rightarrow 1_D(y)$ is not Lebesgue measurable.

but f is measurable, g^{-1} is continuous $\Rightarrow$ this is a counter example.

10pts

10. <u>Proof</u>: Let $C = \{x \in \mathbb{R} : \lim_{n\to\infty} f_n(x) \text{ exists}\}$.

$$= \{x \in \mathbb{R} : \forall k \in \mathbb{N}, \exists N \in \mathbb{N}, \text{ s.t. } \forall m,n > N, |f_m(x) - f_n(x)| < \tfrac{1}{k}\}.$$

$$= \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcap_{n=1}^{\infty} \underbrace{\{x \in \mathbb{R} : |f_m(x) - f_n(x)| < \tfrac{1}{k}\}}_{=: A_{m,n,k}}.$$

Since f_m, f_n are both continuous, $\underbrace{|f_m(\cdot) - f_n(\cdot)|}_{=: h_{n,m}(\cdot)}$ is also continuous.

$\Rightarrow A_{m,n,k} = h_{n,m}^{-1}(\underbrace{(-\infty, \tfrac{1}{k})}_{\text{open}})$. $\Rightarrow A_{m,n,k}$ is open.

$$\Rightarrow C = \underbrace{\bigcap_{k=1}^{\infty} \underbrace{\bigcup_{N=1}^{\infty} \underbrace{\bigcap_{m=1}^{\infty} \underbrace{\bigcap_{n=1}^{\infty} \underbrace{A_{m,n,k}}_{\text{open}}}_{\text{Borel}}}_{\text{Borel}}}_{\text{Borel}}}_{\text{Borel}} \text{ is Borel.}$$

$\square$.

10pts